

TOPIC 6

Cable Management System

This is simply a system of supporting cables. Most common are:-

- Hot
Cooled
- (a) Conduits
 - (b) Bunking
 - (c) Ducting
 - (d) Tray
 - (e) Ladders
 - (f) Basket
 - (g) Modules

1) Conduits

⇒ A tube or pipe in which conductors run;
In effect, the Conduit replaces the PVC sheathing of a cable so that it provides the necessary mechanical protection.

⇒ 3-types of Conduits;

(a) Metal Conduits.

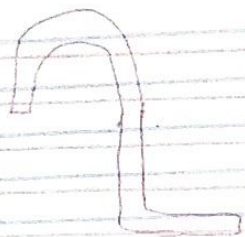
Used in Low Voltage installations.

Mostly heavy gauge welded or 'Solid drawn'.

Heavy gauge is made from a single sheet of steel welded along the seam while Solid drawn is produced in tubular form & is therefore, seamless.

Can be threaded & bent, hence are versatile.
Supplied together with fittings to accomplish installation.

Some typical bends are shown below;
(Using the pipe bending machine).



Swan Neck

(Horizontal to Vertical)



Right angle



Offset

(To enter meter box & termination boxes)

→ For jointing & termination, Fittings are used as shall be discussed later.

Fixing Conduits

→ To Secure or fix Conduits;

- (i) Crampets are used to Secure conduits in place prior to plastering.
- (ii) Saddles are then used hold the the Conduits firmly to the wall.



Crampets

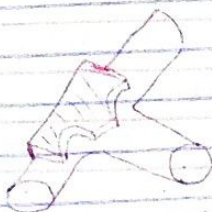


Saddle

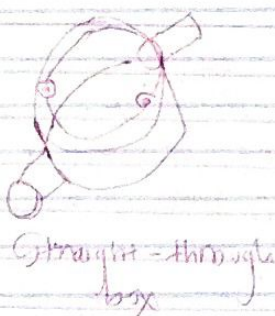
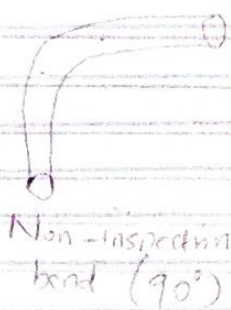
→ Other accessories for use with Conduits include;



Inspection elbow

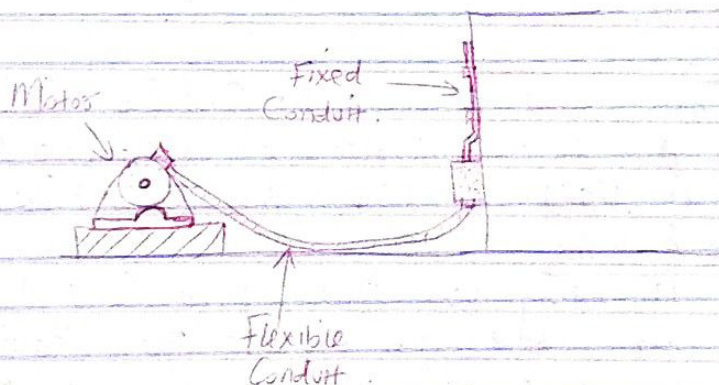


Inspection tee



(b) Flexible Conduit

Flexible Conduit is applied to Supply Machinery. This allows flexibility of movement, allows for vibrations of motor and the motor. Can be easily be aligned or removed for repair.



(c) Non-metallic (PVC) Conduit

- PVC Conduit is suitable for installation system in light industrial premises or offices where surface wiring is required.
- Fittings are identical to those of metallic conduits with the exception that the system is connected by adhesive & not screwing.

Fittings of ~~PVE~~ Conduits

(a) Bends

- These are already pre-fabricated bends to be used with conduits.

N/B Avoid many bends in your installation as will cost money, time and make installation weaker.

(b) Coupling

- They are inter-link between conduits & conduit bends. They have female threads.

(c) Drains

- Where conduits are prone to water moisture, drains are used to draw the water away from the inside. (Very rare scenario).

(d) Bushings & Entry Sprouts

- Bushings & locknuts are mostly used at the termination point to electrical boxes for proper termination.

(e) Unions

- They are used where 2 conduits cannot connect due to unavoidable circumstances hence a union is fitted to link the 2 conduits.

(f) Nipples & hubs

- Where threading the end of a conduit is difficult, nipples are used to provide the required thread.
- Hubs are used to connect conduits to bus boxes or enclosures that don't have a factory threaded entry.

(g) Reducers & reducing washers

- Where conduits come in different sizes, reducers are used to offer a point of connection. Reducing washers serve same purpose as the hubs for conduits of different sizes.

- Fittings can be by hand using the bending spring.
- Nowadays P.V.C Conduits come with fittings (to be covered).
- (In the book notes)

(h) Strap & Clamps

- Used to fasten conduits to fixed structures, beams, hangers etc to prevent sagging.

2) Cable trays

- A Cable tray is used to support insulated electrical cables for power distribution or control or even communication cables.

They are used as an alternative for conduit pipes in a situation whereby:

(a) Large amount of cables passing at a point where they all can't fit into the conduit.

(b) Change in wiring system is anticipated in future hence to re-route is easy when in cable trays than in embedded conduits.

They are commonly used in commercial & industrial wiring systems.

They are made of galvanized steel, stainless steel, aluminium or glass fiber reinforced plastic.

They vary in different types:

- (a) Solid bottom tray - Comes with an enclosure lid and are continuous in length. (Horizontal run-way)
- (b) Ventilated tray - Has openings on the bottom of the tray allowing air circulation, moisture drainage or dust fall-off. (Run-way)
- (c) Ladder tray - Has the cables supported by transverse ladder like bars at regular intervals. Are suited for vertical run-ways.
- (d) Channel tray - Are smaller in size than the others and are used where cables branch off from the main cable tray or to drop cables to a specific point.

3) Cable ducts.

- These are smaller in size than the cable trays and are used to protect cables from external physical interference where the cables are exposed to human activities e.g. in the workshop, office etc.

4) Cable trunkings.

- This is an enclosure, usually with a rectangular cross-section area and with one removable hinged side, that is used to protect cables and provide space for other electrical accessories.
- They protect the cables from dust or water moisture or spills.

A/B

⇒ Cable troughs, ducts or trays are not encouraged where there is a lot of external activities / movement of human and machinery.

- For such a scenario, it is encouraged to embed conduits in concrete walls and run cables through them instead.

5) Ladder.

Similar to Cable tray but in the form of a Ladder; Used in Commercial & industrial installations. It is easy and versatile to install and has good ventilation.

6) Wire basket.

- This is another variation of tray or ladder and is used generally for carrying many small cables such as would be found in data processing installations.

7) Modular.

- This is pre-fabricated system used in Commercial and industrial installations. It comprises distribution boards, special connection units, outlet boxes and pre-set flexible cable lengths that have plugs on each end.

- It requires little effort to erect, the installer just fixes the various connection and outlet units and plugs the cable in at both ends. This clearly avoids the need

WIRING ACCESSORIES

Any device or equipment associated with wiring and are not consumed during the installation process.

① Switches

- A mechanical device for breaking current flow non-automatically at user's convenience.
- Switches installation follow I.E.E guidelines so as to offer convenience & safety to the user.

② Lamp holders

- They hold the bulb in from metallic contact to prevent overheating.
- They are fitted with switches which control the lighting within the same room.

③ Ceiling roses

- They are fitted with terminal intended for connection to a flexible cord. They act as the termination point for lighting installation.

④ Socket Outlets

- A device with protected current-carrying contacts, fixed on a wiring installation to extend a power source to a convenient or required location.

⑤ Regulation Fuses

- A device for opening a circuit by means of a conductor designed to melt when an excessive current flows.

⑥ Circuit breakers

- Used to offer ~~mechanical~~ ^{Load} protection against over-current in the event of a fault. They

mechanically trip breaking off power connection and can re-connect either automatically (ix-cet breakers) or manually (D.B. cct-breakers).

7) Distribution board.

Houses the circuit breakers for the various final circuits. Merely installed inside the house for user convenience in switching on/off (modern).

8) C.C.U.

House the Main Switch which controls power the entire establishment.

9) Junction box.

An accessory that offers easy joining of conductors during installation and becomes points of checking during installation or verification of installation.